

Determining the resilience of grasslands to invasive species

Using Cellular Automata

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Semester 2 project

Interaction between model, theory, experiment, and
simulation in natural sciences



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Abstract

The impact of invasive species has been widely documented, with extreme consequences in terms of biodiversity. With this model we want to determine if an existing ecosystem can maintain its integrity when an invasive species is introduced. We are focusing on danish grasslands, categorized as 6210 by the **Danish Nature Agency (Naturstyrelsen)**, defined by its dependency on animal grazing to not revert into forests, alongside a **balanced ph level**.

Keywords

Grasslands, Denmark, Cellular Automata, Biodiversity, Model, Invasive species

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1 Introduction

The project began with a broad exploration of ecosystem dynamics, aiming to understand how the various components of ecosystems interact over time. Initially, our focus included both invasive plant and animal species, with the goal of building a general model of how invasive organisms interact and reshape ecosystems. However, we quickly realized that trying to model such a wide range of species types introduced excessive complexity. The number of species present in the soil, various trophic levels, and mobility complexities made it difficult to design a well-structured and unified model within the duration of the project. Hence, we redefined our scope to focus exclusively on the conflict between native and invasive plant species. Although these plant species are still pretty complex, they can be modeled more cohesively within a structured ecological framework. The effect of animals on the plant life cycle will be taken into account.^[15]

Invasive plant species pose a growing threat to native ecosystems around the world, altering nutrient cycles, displacing native flora, and reducing biodiversity.^[28] Understanding and predicting their impact is critical for effective ecological management.^[26] This project explores the use of cellular automata (CA) in simulating dynamic invasion processes, focusing on seed dispersion, competitive interactions, and spatial colonisation patterns.^[21]

Cellular automata provides a powerful framework for modelling ecological processes because it can capture the discrete nature of plant community dynamics. As demonstrated by Matsinos and Troumbis (2002),^[26] CA models excel at integrating multiple ecological processes—including species competition, differential dispersal strategies, and disturbance responses—within a single spatially-explicit framework. Their work demonstrated that plants that spread their seeds farther can successfully establish in new areas even when they are weaker competitors, especially when disturbances create gaps in the landscape.

By converting ecological theory into computational principles and rules, the model aims to simulate how invasive species spread and interact with their environment over time.

To create an accurate model, we first need to define the type of area we are working with. In this case the **Danish Nature Agency (Naturstyrelsen)** classifies 'Overdrev' grassland into 3 types:

Category	Characteristics
6120	Very dry grassland or slope vegetation on calcareous sand
6210	Grassland and scrub on more or less calcareous bottom (important orchid localities)
6230	Species-rich grasslands or grasslands on more or less acidic soil

Our group has chosen to go with type 2: 6210.

Each area has its own unique flora, but overall, these grasslands are species-rich and diverse, with many plants that are rare elsewhere in Denmark.^[8]

Our motivation for this choice will be described in detail in section **2.1 How to define ecosystem resilience** as 'Overdrev' grassland have many interesting dynamics.

1.1 Research question

Is it possible to model the spread and resilience of invasive species in a specific type of grassland using cellular automata?^[21]

1.2 Hypothesis

Ecosystems with high sensitivity to invasive species will demonstrate rapid population growth of introduced species across multiple randomized simulation trials, indicating competitive advantages in seed dispersal, resource utilization, nutritional efficiency, or environmental adaptations. Conversely, stable ecosystems resistant to invasion will show consistent decline of introduced species populations toward zero across repeated trials, suggesting native species maintain sufficient competitive advantages to exclude invasive populations.

The cellular automaton simulation will reveal whether the modelled 'overdrev' ecosystem exhibits vulnerability or resistance to our choice of invasive species - *Rosa Rugosa* (described in detail in subsection **3.4.1 Our Invasive species - Rosa Rugosa**) - through consistent population dynamics patterns across multiple randomised test scenarios.

1.3 Connection to semester theme

Our project directly integrates this semester's theme of model, theory, experiment, and simulation in natural sciences. The cellular automaton serves as the methodological bridge connecting all four components.

Theory: We conducted extensive research on plant ecology, species interactions, and invasion dynamics, documenting nutritional requirements, reproductive strategies, and competitive mechanisms.

Model: Based on our theoretical foundation, we developed a conceptual framework predicting invasive species spread patterns and ecosystem responses under various scenarios.

Experiment/ Simulation: The cellular automaton functions as both our experimental apparatus and simulation tool, allowing systematic testing of invasion hypotheses in controlled digital environments.

This integrated approach demonstrates how computational simulations can effectively complement traditional empirical work and theoretical modelling. The cellular automaton creates a simplified yet scientifically grounded experimental environment for testing ecological questions that would be impractical to study through field experimentation alone, providing proof of concept for using computational methods to advance understanding of complex ecological processes.

2 Theory

To begin this project, we first needed to decide on an area to investigate. We wanted to explore a type of grassland where we could observe how invasive species affect an existing ecosystem and its resilience.^[28]

Before moving on, we need to establish the distinction between a native, naturalised, and invasive plant species:^[15]

Native species: These plants originated in the region they currently inhabit. In our case, that would mean the plants that have originated in Denmark/Europe and continue to thrive in the local natural environments to this day.

Naturalized species (non-native): Also known as non-native species, these plants originated in regions different from where they currently grow. A plant is considered naturalized when it has successfully survived and reproduced independently for multiple generations. Denmark hosts numerous naturalized plants, many introduced due to the fact that people like having flowers in their gardens, which are often purchased at nurseries that import them from foreign countries.^[28]

Invasive species: a naturalized plant can also become invasive when it causes harm to the environment or threatens native flora. Invasive plants can inflict substantial and sometimes irreversible damage to ecosystems through various mechanisms, such as:

- Outcompeting native plants for resources^[26]
- Thriving due to absence of natural predators^[15]
- Disrupting established ecosystem, dynamics and food webs
- Damaging agricultural crops
- Overwhelming water bodies, and rendering them unsuitable for native species

Although only approximately 1% of naturalised plants develop into harmful invasive species, those that do establish themselves can prove extremely difficult to contain or eradicate.^[28]

2.1 How to define ecosystem resilience:

The resilience of an ecosystem is defined by the capacity of an ecosystem to return to its original state after being disturbed.^[19] A resilient and healthy ecosystem can take a certain amount of disturbance, depending on the state of the ecosystem, and bounce back to its original state over a period. If the ecosystem is resilient enough, invasive species will have a hard time taking hold in the environment.^[26] To properly observe the effects of our simulation, we wanted to work with a more complex ecosystem where the impact of invasive species would be more visible. After some research, we discovered one of the most biodiverse grassland ecosystems.

We were led to 'overdrevs' a unique type of grassland specific to Denmark that is important due to its high biodiversity.^[8] Over half of all butterfly species in Denmark are found in overdrevs, along with many reptiles that depend on these areas, not to mention the diverse plant life.

Overdrevs were primarily created by humans. Most of these areas were originally large forests that were deforested to create open spaces for animal grazing. These areas are nutrient-poor and dry, with soil often containing sand, gravel, and stones. This made them unsuitable for farming, but as many animals were stabled in Denmark, these grazing fields covered most of the country by the 1800s. Today, as farming and grazing activities have decreased, overdrevs cover only about 0.8% of Danish soil. The

remaining overdrevs are often on uneven ground where other land uses are limited. Due to the limited nutrition in the soil, the recycling of animal feces from grazers is a crucial feature, returning nutrients to the soil and dispersing seeds.

Overdrevs would revert to forests without grazers to keep larger plants down, which is why trees and bushes are sparse in these areas. Grazers are key to maintaining overdrevs and their biodiversity, so today almost all these areas are protected and subject to strict management rules. Large animals are introduced as grazers, while smaller mammals and birds typically inhabit neighbouring habitats rather than the overdrevs themselves.

There are commonly three types of these grasslands/overdrevs:

- 6120: Very dry grassland or slope vegetation on calcareous sand.
- 6210: Grassland and scrub on more or less calcareous bottom (important orchid localities).
- 6230: Species-rich grasslands on more or less acidic soil.

Our group has chosen to focus on type 2: 6210.

We selected this type for two main reasons:

- Its soil pH is balanced between basic and acidic.
- It is home to some of Denmark's rarest plants, which need protection to avoid extinction.

The pH of the soil influences which plant species can thrive in a given area. We chose to go with a neutral pH because it accommodates a wider range of plant species. While some plants can only survive in either acidic or basic conditions, a few have adapted to tolerate both environments. This neutral pH, however, creates a vulnerability because invasive species have a greater opportunity to establish themselves since the soil can support plants with both slightly acidic and basic preferences. The "overdrev" type 2 (6210) ecosystem offers the greatest plant diversity and represents one of Denmark's most sensitive habitats regarding biodiversity and rare plant protection.

Furthermore, Danish "overdrev" ecosystems are characterized by exceptionally dry, nutrient-poor conditions. In these habitats, even minor environmental changes can have widespread impacts. Native plants exist in a constant state of competition for limited resources, making these areas particularly vulnerable to invasive species. Such introductions could be catastrophic, altering the fundamental conditions that define an "overdrev" and negatively affecting the ecosystem's overall biodiversity and health.

After settling on the type of area and understanding its conditions, we narrowed our focus further. We decided to concentrate on north/west Sjælland, as these areas are easily accessible for observations. In Kalundborg, specifically at Røsnæsspiden, there is a type 6210 overdrev divided into ten smaller areas. These areas were established at different times: the oldest in 1970, some newer ones in 1994, and the most recent in 2007. Each area has its own unique flora, but overall, the region is species-rich and diverse, with many plants that are rare elsewhere in Denmark, even on other overdrevs.

These areas are in good health for an overdrev, suggesting that invasive species may struggle to interfere with the ecosystem. Generally, healthier ecosystems are more resilient; therefore, we believe this area is an excellent example for our study, and our findings will likely reflect real-life conditions.

When the overdrev was initially created, cows were used to graze the area. In 2015, goats were introduced to replace cows, as the cows could not effectively control bush growth. Later, both donkeys and horses were also introduced for grazing, but now it's primarily goats that maintain this overdrev. These goats are efficient at preserving the grassland and play a crucial role in recycling nutrients back into the soil. Additionally, some plant species are dispersed through their fur or digestive tracts.

3 Model

3.1 Theoretical Model

Our simulation uses the ecological theory to model plant community dynamics in spatially varied environments, specifically focusing on invasive species spread in Danish coastal grasslands.^{[21] [6]} We combined several theoretical frameworks to develop a plant competition and coexistence model.^[16]

The model is based on resource competition theory, where plants compete for three essential resources: water, nitrogen, and phosphorous.^[26] Each species has different resource requirements, which can allow for niche differentiation. Additionally, we also model dynamic plant lifecycles- including growth, reproduction, and mortality rather than treating plants as unchanging or static entities. This helps us examine how species with different growth strategies respond to environmental changes.^[18] The model incorporates niche differentiation, which refers to how different species can coexist in the same area by specializing in different resource use patterns, timing of resource uptake, or spatial resource acquisition, thereby reducing direct competition.^[16]

3.2 Key ecological processes

Resource dynamics

Resources form the foundation of our model. Rather than assuming uniform resource distribution, we created spatially heterogeneous landscapes where resources vary across the grid. Precipitation events provide stochastic resource inputs with variable intensity affecting different-sized areas, creating shifting patches of high resource availability. Plants consume resources according to their species-specific needs, creating local depletion around established individuals.^[23]

A key feature is the decomposition feedback loop: when plants die, they return nutrients to the soil, creating spatial legacies where previous vegetation influences current resource patterns. Natural regeneration processes provide background nutrient cycling that slowly replenishes resources

Plant life cycle

We represent the complete plant lifecycle through five discrete stages: seed, seedling, juvenile, mature, flowering, and death. Each stage has different resource requirements, so environmental stress affects plants differently depending on their developmental stage.

Seeds require specific conditions to germinate—insufficient resources keep them dormant rather than killing them. Seedlings and juveniles have high resource demands relative to their size, making them vulnerable to competition. Mature plants survive resource shortages better but need adequate resources to transition to flowering.

The flowering stage drives reproduction, but we constrained it seasonally. Each species has a specific flowering window, so even resource-rich plants can't reproduce year-round. This temporal niche differentiation helps explain how species with similar resource requirements can coexist.

Competition

When multiple seeds compete for the same location, we use a probability-based system that considers both species fitness and propagule pressure:

Species Score = (Water compatibility + Nitrogen compatibility + Phosphorus compatibility) × Number of seeds

This approach captures both the advantage of producing many seeds and the importance of environmental suitability.

Dispersal mechanisms

Our model incorporates 3 distinct dispersal strategies:

Wind dispersal: Seeds follow directional wind patterns with daily variation in wind direction and strength. We implemented a simple but effective wind system where wind direction cycles through eight compass directions (N, NE, E, SE, S, SW, W, NW) over an 8-day period. Wind strength varies randomly each day between 50% and 200% of base strength.

When a flowering plant produces seeds, each seed follows the daily wind direction but with added turbulence - we add random variation of ± 2 grid cells to both X and Y coordinates to simulate air currents and seed aerodynamics. The final dispersal distance gets multiplied by the species' dispersal bonus, so species with better wind-adapted seeds (higher dispersal bonus values) travel further.

Animal dispersal: Seeds are transported by animals in random directions with moderate dispersal distances. We simplified animal behaviour by assuming animals don't preferentially carry seeds to favourable habitats (though this would be more realistic). Instead, each seed gets deposited in a random direction at a distance determined by the species' dispersal range.

This mechanism creates the patchy, unpredictable distributions characteristic of animal-dispersed plants.

Underground shooting: This represents vegetative reproduction through rhizomes, runners, or root sprouting. The parent plant produces clonal offspring in one of the eight immediately adjacent cells, chosen randomly. Unlike seed dispersal, this method creates new individuals directly as seedlings (bypassing the seed stage entirely) and only works if the target cell is empty.

Each dispersal mechanism produces distinctive spatial signatures in our simulation results, allowing us to test how different dispersal strategies perform under varying environmental conditions.

Seasonal phenology

Different species flowers at different times of year, from early spring through late fall. This temporal separation reduces direct competition for reproductive resources, which is not explicitly modelled. Species with early flowering periods can reproduce before summer drought stress, while late-flowering species can take advantage of autumn rains.

3.3 Mathematical Representation

Rather than developing complex mathematical equations, we chose to implement our ecological processes through computational rules that are easier to understand and modify. However, the core relationships can still be expressed mathematically to show how the different components interact.

Resource dynamics

The basic resource equation is as follows:

$$\begin{aligned} \textit{Tomorrow's Resources} &= \textit{Today's Resources} + \textit{Inputs Consumption} + \textit{Decomposition} \\ R_{y,x,t+1} &= R_{y,x,t} + I_{y,x,t} - C_{y,x,t} + D_{y,x,t} \end{aligned}$$

where:

y = row position (y coordinate)

x = column position (x coordinate)

t = time (today)

t + 1 = tomorrow (the next day)

$R_{y,x,t}$ = Resources at location (y,x) on day t

$R_{y,x,t+1}$ = Resources at that same location (y,x) on day t+1

Plant Growth

Plant growth depends on whether local resources exceed the species' requirement's:

Plant advances stage if the available resources $\geq$ Species requirements

This binary approach matches biological reality since plants either have sufficient resources to grow or they don't.

Plant Stages

The plant life cycle is divided into 6 stages:

Seed, Seedling, Juvenile, Mature, Flowering and Death.

We wanted to incorporate two different life cycles in the simulation, annual and perennial:

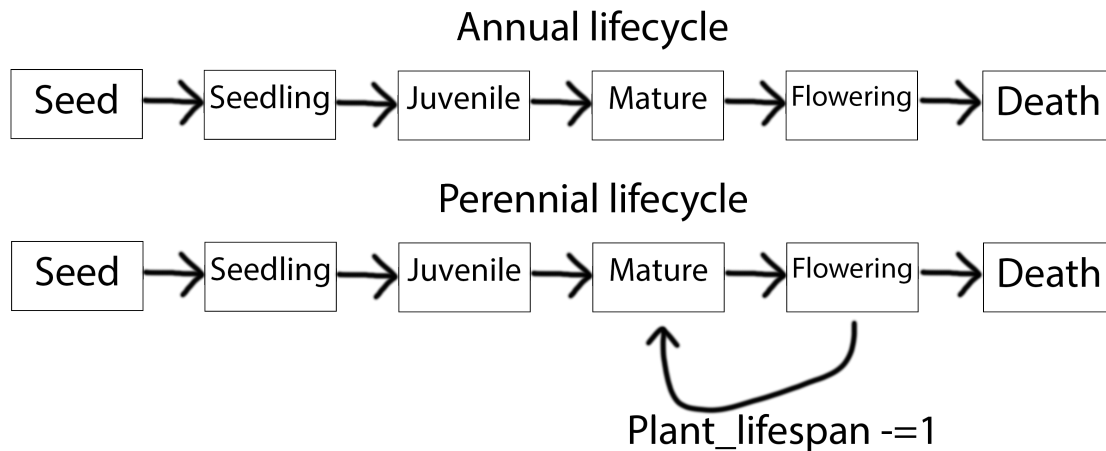


Figure 1: Representation of the 2 lifecycles

As seen in Figure 1, the annual life cycle would follow the 6 stages, as the plant progresses in age, until the final stage: Death.

The perennial life cycle would follow the first 5 stages, but in the end of the flowering period, instead of entering Death stage, the plant regresses to the Mature stage. The year after the perennial plant enters into Flowering once again before regression to Mature stage.

The plants progressing to Death stage, or regressing to the Mature stage follows the expression:

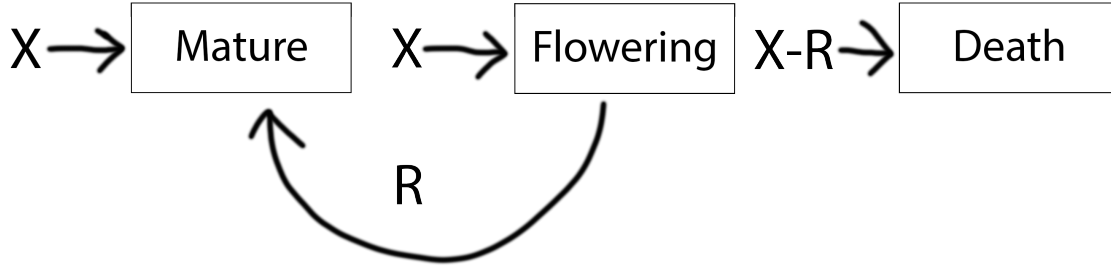


Figure 2: X is the number of plants in the stage, R are the number of perennial plants regressing

Competition When multiple seeds from different species land in the same location, we determine the winner through a probability-based competition system that accounts for both environmental suitability and propagule pressure. The competition process follows these steps:

1. Calculate species score: For each species with seeds in the location, we evaluate how well suited they are to the local environmental conditions:

$$\text{Species Score} = (\text{Water Score} + \text{Nitrogen Score} + \text{Phosphorus Score}) \times \text{Number of Seeds}$$

$$P_{score} = (P_{water} + P_N + P_P) * \text{Number of seeds}$$

Where each resource score reflects how well the species' requirement match available resources (scored 1-5 based on sufficiency thresholds.)

2. Calculate total competition: Sum all species scores competing for the location:

$$\text{Total score} = \text{Species Score}_1 + \text{Species Score}_2 + \text{Species Score}_3 + \text{Species Score}_n$$

3. Determine establishment probability: Each species' chance of winning the location:

$$\text{Establishment probability} = \text{Species Score} / \text{Total Score}$$

4. Select winner: generate a random number (0-1) and compare against cumulative probabilities to determine which species successfully establishes.

This approach captures 2 key ecological principles: species with better environmental matching have higher establishment success, and species producing more seeds (higher propagule pressure) gain competitive advantages through sheer numbers.

3.4 Plant information

Our simulation requires comprehensive data on native, naturalised, and invasive plant species to accurately model ecosystem dynamics. We developed an extensive database containing detailed information about species present in the Rødnaes "Overdrev" ecosystem, enabling us to create a realistic simulation framework.

Database Parameters Our research identified approximately 150 species present at Rødnaes "overdrev." For each species, we collected the following critical data points:

- Species name
- Seed dispersal mechanisms and movement patterns
- Annual seed production
- Growth characteristics and maximum size
- Water requirements
- Nitrogen requirements
- Phosphorus requirements
- Life cycle classification (annual/perennial)
- Regional abundance
- Origin status (naturally occurring/ adapted/ invasive)
- Palatability to grazing animals (goats)

Nutritional scoring system We implemented a standardized scoring system for resource requirements, where species needs are rated from 1-5:

- Very low needs: 1
- Low needs: 2
- Moderate needs: 3
- High needs: 4
- Very high needs: 5

The following are a few examples of our species:

Species 1. *Achillea millefolium* (Common Yarrow):



Figure 3: (Picture from Naturbasen, Photograph by Thomas Hansen, Taken: 30/7-2024)

Reproduction and dispersal: Common Yarrow flowers from June through September, producing wind-dispersed seeds. The species also propagates vegetatively through underground shoots, typically spreading 3-10 inches annually. Under undisturbed conditions, approximately 75% of seeds successfully germinate, though rates commonly range between 25-50%.

Seed Production: Each plant produces approximately 1,600 seeds annually.

Growth Characteristics: Plants reach heights up to 60 cm, completing their primary growth cycle in approximately four months.

Resource Requirements:

- Water needs: Low(2)
- Nitrogen needs: Low(2)
- Phosphorous needs: Moderate (3)

Life cycle: Perennial species with individual plants typically surviving 2-5 years.

Distribution: Very common throughout Denmark, found in diverse habitats.

Classification: Naturally occurring native species.

Grazing Pressure: Readily consumed by goats, comprising up to 40% of their diet in some areas where Common Yarrow is abundant.

Species 2. Asparagus Officinalis (Asparagus):



Figure 4: (Picture from Naturbasen, Photograph by Henrik Baark, Taken: 1/9-2024)

Reproduction and dispersal: Asparagus flowers from June through August, producing insect pollinated berries. Birds eat the berries and disperse seeds through their droppings. Under optimum conditions approximately 75% of the seeds successfully germinate.

Seed Production: Individual plants produce berries containing approximately 5 black seeds each. Berry production correlates with plant maturity: first-year plants produce no berries, second-year plants yield 5-6 berries (25-30 seeds), while mature specimens can produce hundreds of berries annually.

Growth Characteristics: Plants reach max heights up to 200 cm, completing their primary growth cycle over a few years before becoming a mature plant. If possible, they prefer lots of sun to grow to maximum heights.

Resource Requirements:

- Water needs: Moderate(3)
- Nitrogen needs: Moderate(3)
- Phosphorous needs: Moderate (3)

Life cycle: Perennial species with individual lifespans of 20-30 years.

Distribution: Nearly endangered in Denmark, found in few habitats.

Classification: Native species naturally occurring in Denmark.

Grazing Pressure: Readily consumed by goats, it's a treat for goats when they find the plant.

Species 3. Brachypodium Pinnatum (Tor-grass):



Figure 5: (Picture from Naturbasen, Photograph by Per Taudal Poulsen, Taken: 25/7-2012)

Reproduction and dispersal: Tor-grass spreads primarily through underground rhizomes, with flowering occurring from June through July. While seeds can attach to animal fur, this dispersal mechanism is negligible; vegetative spread dominates reproduction.

Seed Production: Expansion occurs exclusively through underground shoots during growing seasons. Although seeds are produced, their minimal dispersal effectiveness renders them insignificant for our simulation model.

Growth Characteristics: Plants reach heights up to 70-120 cm.

Resource Requirements:

- Water needs: Low(2)
- Nitrogen needs: High(4)
- Phosphorous needs: Moderate (3)

Life cycle: Perennial species with root systems potentially surviving over 300 years.

Distribution: Almost endangered throughout Denmark, found in a few remaining suitable habitats.

Classification: Native species naturally occurring in Denmark.

Grazing Pressure: Readily consumed by goats; grazing pressure effectively controls spread. Without grazing management, this species would exhibit exponential expansion.

Species 4. Bromus Hordeaceus ssp. Hordeaceus (Soft chess):



Figure 6: (Soft chess , 25-05-2024, Photo: Lars Bindholt)

Reproduction and dispersal: Soft chess flowers from April through July, producing seeds dispersed by wind and animals. Human activities, including machinery and soil transport, also facilitate dispersal.

Seed Production: Each plant generates approximately 1,800 seeds annually; however, germination rates are extremely low.

Growth Characteristics: Plants reach heights up to 20-70 cm.

Resource Requirements:

- Water needs: Moderate(3)
- Nitrogen needs: Moderate(3)
- Phosphorous needs: Moderate (3)

Life cycle: Annual or bi-annual species, with life cycle duration dependent on environmental conditions.

Distribution: Common throughout Denmark, particularly found in habitats like meadows and roadsides habitats.

Classification: Native species naturally occurring in Denmark.

Grazing Pressure: Readily consumed by goats, as a type of grass, representing a significant component of their diet.

Species 5. Carlina Vulgaris (Carline Thistle):



Figure 7: (Picture from Naturbasen, Photograph by Søren Nygaard, Taken: 14/11-2024)

Reproduction and dispersal: Carline Thistle flowers from July through September, relying on insect pollination. Wind-dispersed seeds feature tufted structures enabling limited dispersal of several meters.

Seed Production: Individual plants exhibit minimal reproductive success and spread capacity.

Growth Characteristics: Plants reach heights up to 50 cm.

Resource Requirements:

- Water needs: Low(2)
- Nitrogen needs: Low(2)
- Phosphorous needs: Low (2)

Life cycle: Perennial species with typical lifespans of 2 years.

Distribution: Nearly endangered throughout Denmark, restricted to few remaining habitats.

Classification: Native species naturally occurring in Denmark.

Grazing Pressure: Can be consumed by goats. Goats don't seek out to this species preferentially, but might consume it, if it is between other species they like. Additionally, it is toxic to other grazing animals except goats.

The following are the example calculation for our invasive species:

Species 6. Rosa Rugosa (Japanese / Beach Rose):



Figure 8: (Photograph by Mette-Marie Lilje Andreasen, Taken: 20/5-2025)

Reproduction and dispersal: Japanese Rose flowers from June through August with beetle pollination. Birds and mammals up to fox size consume the numerous fruits. Seeds maintain viability while floating in salt or fresh water for up to one year. Underground shoots extending 2-3 meters provide additional vegetative reproduction, creating robust dispersal mechanisms.

Seed Production: Mature plants produce 600-1,300 seeds per square meter annually. Extensive underground shoots of 2-3 meters create clonal colonies through vegetative reproduction.

Growth Characteristics: Plants range from 20-200 cm in height, forming dense thickets that alter local environments. Originally planted on coastlines for sand stabilization, these bushes create monocultures that shade out native vegetation.

Reproduction and dispersal: Japanese Rose flowers from June through August with beetle pollination. Birds and mammals up to fox size consume the numerous fruits. Seeds maintain viability while floating in salt or fresh water for up to one year. Underground shoots extending 2-3 meters provide additional vegetative reproduction, creating robust dispersal mechanisms.

Resource Requirements:

- Water needs: Very Low(1)
- Nitrogen needs: Moderate(3)
- Phosphorous needs: High (4)

Life cycle: Perennial species surviving 20-50 years, typically reaching reproductive maturity within 3-5 years.

Distribution: Common throughout Denmark across diverse habitats.

Classification: Invasive species - native to eastern Asia. Denmark's most problematic invasive plant, catastrophic for native flora. Imported in the 1800s with wild populations first recorded in 1875. Currently found in hedgerows, summer house areas, and coastal zones.

Grazing Pressure: Can be consumed by goats but is not the preferred plant. Goats represent the only grazing animals capable of sufficiently intensive browsing to control the Rosa Rugosa spread.

3.4.1 Our Invasive species - Rosa Rugosa

Environmental Impact

Rosa Rugosa serves as the invasive species in our simulation, representing Denmark's most widespread invasive plant. This species poses a significant threat to coastal ecosystems throughout the country.

While *Rosa Rugosa* effectively stabilizes sand with its extensive root system and prevents erosion, which was originally one of the reasons for its introduction, it fundamentally alters surrounding environments and eliminates native plant species. Since Denmark's coastal habitats represent some of the country's most pristine native ecosystems, *Rosa Rugosa*'s presence creates substantial ecological concerns.

Historical Introduction and spread

Rosa Rugosa originates from Northeast Asia and arrived in Europe during the 1800's with the first wild populations recorded in 1875. The species gained significant momentum in Denmark during the 1950s housing boom, when it became a popular choice for summerhouse gardens. Its appeal stemmed from multiple attractive characteristics: effective erosion control, edible fruits suitable for jams, beautiful fragrant pink flowers, and low maintenance requirements. The Danish government further accelerated its spread by using *Rosa Rugosa* as an inexpensive windbreak solution, inadvertently facilitating its establishment across the landscape.

Ecological threats

Rosa Rugosa's primary danger lies in its ability to form dense thickets that shade and eliminate surrounding vegetation. This aggressive growth pattern allows it to completely overtake native flora, creating monocultures with severely reduced biodiversity. The resulting environment supports limited fauna, disrupting entire ecosystem food webs and ecological relationships.

Management challenges

Rosa Rugosa's resilience makes eradication extremely difficult. Complete removal requires excavating every root shoot; otherwise, the plant regenerates and continues spreading. Limited control methods include intensive goat grazing—the only herbivore capable of suppressing the species—or strong pesticide applications. The plant's exceptional hardiness and seed dormancy capabilities, which can last multiple years, have earned it the distinction of Denmark's most problematic invasive species.

Reproduction and dispersal mechanisms

Rosa Rugosa employs multiple reproductive strategies for rapid colonization. Each fruit contains 20-120 seeds, with mature plants producing 600-1,300 seeds per square meter annually. Seeds maintain viability for years in both soil and water, while fruits dispersed by wildlife extend the species' range. The extensive root system, typically reaching 0.5-1 meter deep, complicates removal efforts and enables vegetative reproduction.

Growth rates are exponential, with populations commonly doubling every three years across Danish locations. The species can hybridize with native roses, creating additional competitive pressure on indigenous flora. Even after plant death, thorns persist on the ground for years, creating lasting habitat modifications.

Due to *Rosa Rugosa*'s extensive establishment throughout Europe and Denmark, complete eradication is economically and practically unfeasible. Current management strategies focus on local control programs rather than total elimination. Success depends heavily on community involvement, particularly in summerhouse areas where the species is most prevalent. Without sustained local engagement, effective population control remains impossible, relegating *Rosa Rugosa* to a species that can be managed locally but only contained at acceptable levels nationally.

3.5 Data Implementation in the simulation

Our model uses the data mentioned in the subsection above, to categorise the species through the following key data categories that drives the simulation:

1. Seed dispersal and movement patterns

Understanding seed dispersal mechanisms is crucial for modeling plant reproduction and spread. We determine seed release timing by tracking flowering periods, as seeds are typically dispersed at the conclusion of each flowering cycle. Each species is assigned one of three dispersal algorithms: wind dispersal follows the daily wind direction vector with species-specific distance multipliers; animal dispersal places seeds randomly within species-defined radii; vegetative reproduction places offspring in adjacent empty cells. Seed release timing is synchronized with each species' flowering period, creating temporal variation in dispersal events across the landscape.

Temporal dispersal patterns significantly influence seed survival rates. Many species release seeds during similar periods, creating concentrated foraging opportunities for birds and small mammals. This temporal overlap results in increased seed predation. However, some species have evolved alternative strategies: seeds adapted for animal gut passage benefit from this interaction, while species flowering outside peak periods experience reduced seed loss to predators.

2. Seed production quantities: Species exhibit remarkable variation in reproductive strategies. Annual seed production values (ranging from 10-1,300 seeds per plant) are used to populate the 3D seed competition array. High-production species like *Rosa rugosa* contribute more seeds to competitive events, while species relying on vegetative reproduction contribute fewer seeds but bypass the vulnerable seed stage entirely. Some produce numerous seeds with low germination rates, while others generate fewer seeds with higher viability. Certain species rely primarily on vegetative reproduction through underground shoots, producing few viable seeds. Understanding these quantitative differences enables accurate modelling of population dynamics and colonisation rates.

3. Plant size and growth characteristics: plant architecture influences both competitive interactions and reproductive success. Size classifications help distinguish between herbaceous plants, shrubs, and trees within our simulation framework. Larger plants cast more shade, affecting competition, and typically achieve greater seed dispersal distances due to increased height and canopy spread.

4. Nutritional requirements: Our simulation incorporates species-specific water, nitrogen, and phosphorus requirements (scored 1-5) and compares it against local resource grid values to determine growth advancement. Plants progress to the next life stage only when all three resource thresholds are met, creating realistic competitive hierarchies based on resource efficiency.

5. Life cycle classification: Distinguishing between annual and perennial species is essential for tracking population dynamics over time. When plants die, their biomass is converted to resource units and added back to the local resource grids, with decomposition rates varying by life cycle type. Modelling natural lifespans enables realistic population turnover and resource recycling within the ecosystem.

6. Species abundance data: Real world abundance patterns serve as calibration targets for our simulation results. We adjust initial population distributions and validate that common species maintain dominance while rare species remain at low frequencies, ensuring model realism.

7. Origin classification: since research objective focuses on ecosystem vulnerability to invasive species, we must categorise each plant as native, naturalised, or invasive. Invasive species can be introduced at specified time points and locations, allowing us to test ecosystem responses under various invasion pressures.

8. Goat Grazing likeability: The Rødne "overdrev" ecosystem experiences active goat grazing, which significantly influences plant community composition. Grazing palatability scores modify plant

survival rates in our simulation. Highly palatable species face increased mortality pressure, reflecting the real grazing management at Rødneæs, while unpalatable species like *Rosa rugosa* gain competitive advantages.

3.5.1 Competition

When analyzing the different plants, we needed to take into consideration why certain species successfully outcompete others in natural environments. Our research identifies two primary competitive advantages that determine species success: nutritional efficiency and reproductive capacity.

Species exhibit varying resource requirements for water, nitrogen, and phosphorus, creating competitive hierarchies based on efficiency. Our data reveals significant differences in total nutritional demands:

- *Achillea millefolium* (Common Yarrow): Total nutritional value of 7
- *Asparagus officinalis* (Asparagus): Total nutritional value of 9
- *Carlina vulgaris* (Carlina Thistle): Total nutritional value of 6

In nutrient-poor "overdrev" environments, species with lower resource requirements generally maintain competitive advantages. *Rosa rugosa* (Japanese Rose), our focal invasive species, demonstrates this principle with a total nutritional value of 6. This relatively low resource demand enables it to outcompete native species with higher nutritional requirements.

Beyond overall nutritional efficiency, specific adaptations enhance competitive ability. *Rosa rugosa* develops extensive root systems reaching depths of one meter, enabling access to deeper water sources and supporting its classification as a low-water-need species. This deep rooting strategy parallels adaptations seen in other drought-tolerant species like asparagus.

Reproductive and dispersal advantages: While nutritional efficiency provides one competitive pathway, reproductive success offers an alternative route to dominance. Comparing *Rosa rugosa* with *Carlina vulgaris* illustrates this principle clearly. Despite similar nutritional requirements (values of 6), their reproductive strategies create vastly different competitive outcomes.

Carlina vulgaris faces severe reproductive limitations: minimal seed production, poor germination rates, and absence of vegetative reproduction. Individual plants typically produce only 1-2 offspring, severely constraining population growth and spatial expansion.

In contrast, *Rosa rugosa* employs multiple reproductive strategies that ensure rapid population expansion:

- Dual reproduction modes: Both sexual (seed) and asexual (vegetative) reproduction
- Clonal growth: Individual plants can double in size over three-year periods
- High seed viability: Approximately 80% germination success
- Exceptional seed production: 600-1,300 seeds per square meter from mature plants
- Seed longevity: Seeds remain viable for multiple years before germination

Competitive success strategies: Our analysis demonstrates that species achieve competitive dominance through two distinct but potentially complementary pathways. First, nutritional efficiency allows species to thrive in resource-limited environments where high-demand competitors struggle. Second, superior reproductive and dispersal capabilities enable rapid population growth and spatial colonization, even when nutritional advantages are minimal.

Rosa rugosa's invasion success exemplifies how combining both strategies creates formidable competitive pressure on native species. Its moderate resource requirements suit the nutrient-poor overdev environment, while its exceptional reproductive capacity enables rapid expansion once established. This dual advantage explains why *Rosa rugosa* has become Denmark's most problematic invasive plant species.

4 Implementation

4.1 System Architecture

We built the simulation using Python with a modular design that separates different system components. This made development easier and allows us to modify individual components without breaking the entire system.

4.1.1 Core Data Structures

The grid class manages all the spatial data using NumPy arrays. We chose NumPy for performance, since just having Python loops over large grids would be significantly slower. The main arrays track species identity, growth stage, and resource levels at each location. We are also using an extra dimension as a seed array for tracking dispersal and competition.

Core Spatial Arrays:

- `Species_grid` (int8): Tracks which species occupies each location (0 = empty, 1-150 = species ID)
- `growth_stage_grid` (int8): Records life cycle stage for each plant (0 = empty, 1 = seed, 2 = seedling, 3 = juvenile, 4 = mature, 5 = flowering)
- `age_grid` (int16): Tracks how long each plant has been in its current growth stage

Resource Arrays:

- `water_grid` (float32): Current water availability at each location
- `nitrogen_grid` (float32): available nitrogen levels
- `phosphorous_grid` (float32): available phosphorous levels
- `decomposition_grid` (float32): nutrient input from decomposing plant matter

Seed Competition Array:

- `seed_array` (3D array, int16): Dimensions are [y, x, species], where each cell contains the number of seeds of each species competing for that location. This 3D structure allows us to efficiently track multi-species competition without nested loops.

Environmental Tracking:

- `Wind_directional_grid` (int8): Current wind direction affecting seed dispersal

As we progressed through the project, we realised the importance of memory management as we scaled to larger grids. Using int8 for categorical data (such as species and growth stages) and float32 for resources reduced our memory usage significantly.

The `PlantSpeciesData` class stores all the species-specific traits. This object-oriented approach makes it easier to add new species or modify the existing ones. Each species has parameters for resource requirements, dispersal characteristics, and life cycle timing.

Resource management happens through a separate `Resource` class that handles environmental dynamics including precipitation events, natural regeneration, and decomposition processes. This separation allows us to experiment with different environmental scenarios without modifying the core plant growth algorithms.

4.1.2 Data export system

Our data export system captures three types of data: the population counts by species and growth stage, spacial distributions of all plants, and resource gradients across the landscape. The export happens automatically at regular intervals of every 14 days to prevent data loss if the simulation crashes. We also aggregate spacial data to reduce file sizes- instead of exporting every cell, we average resources and count plants within larger blocks.

4.2 Core Algorithms

4.2.1 Resource Dynamics

Since precipitation drives most of the interesting resource dynamics, our rain algorithms create circular precipitation events with random centres, radii, and intensities: The spatial pattern emerges from ap-

```
def rain(grid, panel_gap, height, width):  
    """Optimized rain simulation"""  
    rain_chance = 0.95  
  
    if np.random.rand() <= rain_chance:  
        radius = np.random.randint(1, 100)  
        center_x = np.random.randint(width) + panel_gap  
        center_y = np.random.randint(height)  
        intensity = int(np.random.rand() * 100)
```

Figure 9: Code snippet from the grid.py file showing the rain algorithm implementation

plying a decay function based on distance from the rain center. We use vectorized NumPy operations to calculate distances for all the grid cells simultaneously, which is much faster than nested loops. Figure 8 above shows the stochastic precipitation system that creates circular rain events with randomized centers, radii (1-100 units), and intensities. The algorithm uses a 95% probability threshold for rain events and applies vectorized NumPy operations to efficiently calculate distance-based resource distribution across the grid, creating realistic spatial autocorrelation in water availability.

This approach creates realistic spatial autocorrelation in water availability- nearby cells tend to have similar water levels, but the pattern shifts over time as new rain events occur.

Background resource regeneration happens more simply through random draws from normal distributions. This represents slow processes like nitrogen fixation that operate continuously but at low rates.

4.2.2 Plant growth

The growth algorithm examines every occupied cell and determines whether each plant can advance to its next developmental stage. This involves checking resource availability against species-specific thresholds:

Figure 10 shows how we implemented this as a series of conditional statements rather than mathematical functions because it's easier to understand and modify. Each growth stage has different requirements, and resource consumption occurs immediately when plants advance to stages.

Seasonal flowering requires additional logic that checks both resource availability and calendar timing. Plants can only transition to flowering during their species-specific flowering window, even if resources are abundant.

```

if stage == 0: # Seed stage
if (water >= species.water_need and
    nitrogen >= species.nitrogen_need and
    phosphorous >= species.phosphorous_need):
    grid.plant_stage[y, x] = 1
    grid.reset_plant_age(y, x)
    germinations += 1

```

Figure 10: Code snippet from main.py showing the plant growth conditional logic

One challenge we encountered was preventing unrealistic oscillation between growth stages. Early versions allowed plants to advance and regress repeatedly as resources fluctuated.

4.2.3 Seed dispersal

Dispersal involves two phases: the seed production by flowering plants and the seed establishment through competition.

The dispersal algorithm first identifies all flowering plants, then distributes their seeds according to species-specific mechanisms. Figure 11 shows how wind dispersal incorporates a daily wind direction that cycles through eight compass directions:

```

# Get current day from main simulation (if not available, use a fallback)
day = getattr(self, 'day', 0)

# Simplified wind direction approach - 8 main directions, cycling clockwise
# 0: North, 1: Northeast, 2: East, 3: Southeast, 4: South,
# 5: Southwest, 6: West, 7: Northwest
daily_wind_direction = (day % 8) # Cycles through 8 directions
wind_strength = np.random.uniform(0.5, 2.0) # Random strength multiplier

# Direction vectors for the 8 main directions (N, NE, E, SE, S, SW, W, NW)
wind_directions = [
    (0, -1), # North: y decreases
    (1, -1), # Northeast
    (1, 0), # East: x increases
    (1, 1), # Southeast
    (0, 1), # South: y increases
    (-1, 1), # Southwest
    (-1, 0), # West: x decreases
    (-1, -1) # Northwest
]

```

Figure 11: Code snippet from grid.py displaying the wind dispersal system

This creates predictable but variable wind patterns. Seeds follow the wind direction but with added randomness to simulate turbulence and variation in seed aerodynamics.

Animal dispersal simply selects random directions for each seed. Although we realize that this is a bit

unrealistic, since a real animal dispersal often shows a directional bias toward favourable habitats, that would be harder to model. However, this approach still captures the key feature of animal dispersal, which is the reduced predictability as compared to wind dispersal.

Vegetative reproduction places offspring in adjacent cells, but only cells are empty. This creates the limited dispersal and dense clustering characteristic of clonal plants.

4.2.4 Seed competition

When multiple species disperse seeds to the same location as shown in figure 12, we resolve the competition through numerical dominance, so whichever species contributes the most seeds win. This ‘majority rule’ approach is simple but captures priority effects and the advantage of high seed production.

```
        if p_scores and sum(p_scores) > 0:
# Calculate sprouting probabilities
total_score = sum(p_scores)
probabilities = [score / total_score for score in p_scores]

# Select winner based on probability
random_value = np.random.random()
cumulative_prob = 0
winning_species = None

for j, prob in enumerate(probabilities):
    cumulative_prob += prob
    if random_value <= cumulative_prob:
        winning_species = species_with_seeds[j]
        break

# Sprout the winning species
if winning_species is not None:
    self.plant_species[y, x] = winning_species
    self.plant_stage[y, x] = 0 # Start as seed (will germinate in plant_growth)
    self.plant_age[y, x] = 0
    sprouted_count += 1
```

Figure 12: Code snippet from grid.py showing the seed competition algorithm

4.3 User Interface

The simulation includes a real-time visualisation system built with Pygame. From figure 13, it can be seen that the plants appear as coloured rectangles, with colour indicating species and brightness indicating growth stage. This provides immediate feedback about the simulation dynamics. A statistics panel shows current population counts, simulation time (day and year), and the system status. An additional control key, the space key, allows the user to run through the simulation manually.

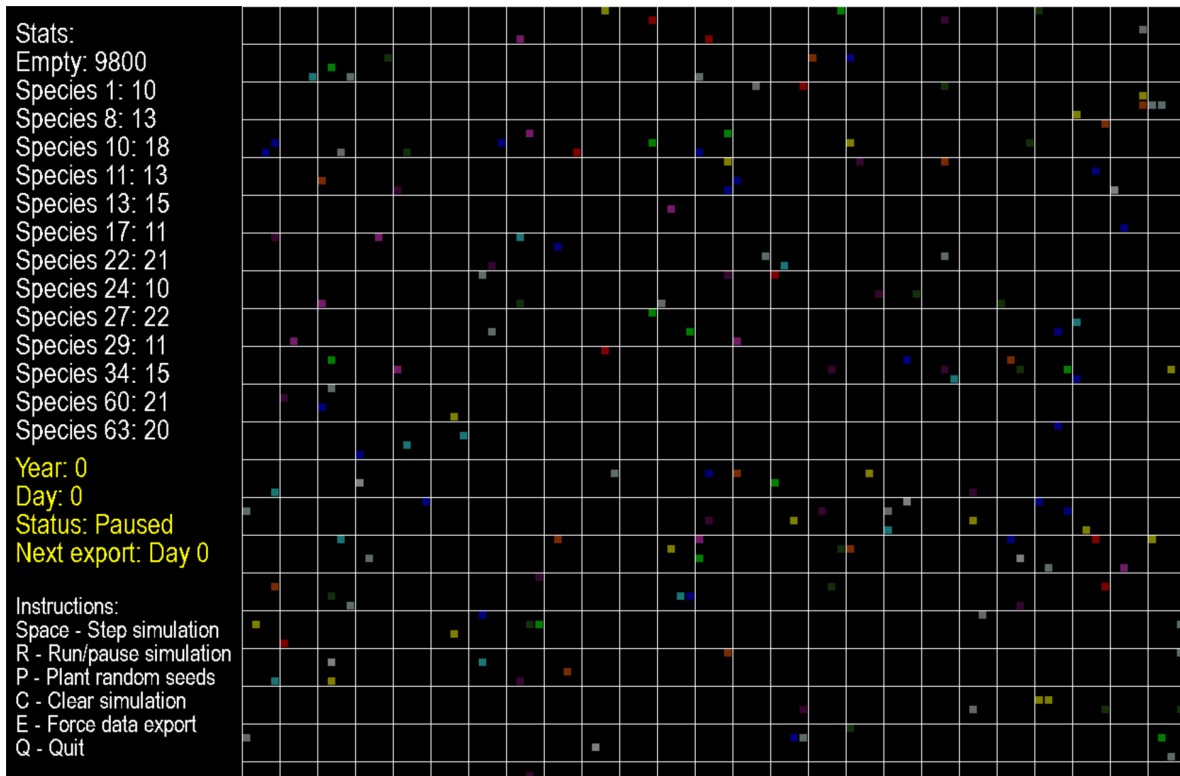


Figure 13: Screenshot from main.py: `updates_stats` function, showing the real-time simulation interface

While primarily designed for research, the visual interface proved valuable for debugging and developing intuition about model behaviour. Watching the simulation run often revealed unexpected patterns that weren't obvious from the data alone.

5 Analysis and Results

The base of our hypothesis relies on population numbers and statistical analysis, following this process:

1. Make sure the simulation works
2. Run the simulation 50-100 times
3. Get the data from each test
4. Compare the population of each species with itself
5. Concentrate that into a mean
6. Compare the mean of each species
7. Compare the graphs of different timesteps

5.1 Resilience metrics

Ecosystem resilience is defined as the capacity of an ecosystem to return to its original state after being disturbed.

To quantify ecosystem resilience objectively, we utilise the following metrics:

1. The proportion of native species with healthy populations left after invasion.
2. The number of time-steps needed for the system to recover to within 10% of pre-invasive states.
3. The negative correlation of invasive species' introduction quantity and establishment success.

Table 1: Test overview

Scenario	Number of species present	Dominant species (Excluding empty tiles)	Expected result	Actual results
Baseline	9.5 (One species was not correctly added)	Asparagus officinalis (Asparges)	Stable ecosystem	Stagnant ecosystem
Higher population	9.5 (One species was not correctly added)	Cichorium Intybus (Cikorie)	Stable ecosystem	Stagnant ecosystem
All 12 species + higher population	12	Brachypodium pinnatum (Bakkestil-laks)	Significant spread of the invasive species	Stagnant ecosystem
Semi fixed seed spreading	12	Rosa Rugosa (Rynket rose)	Significant spread of the invasive species	Significant spread of the invasive species

5.2 Testing

Test 1

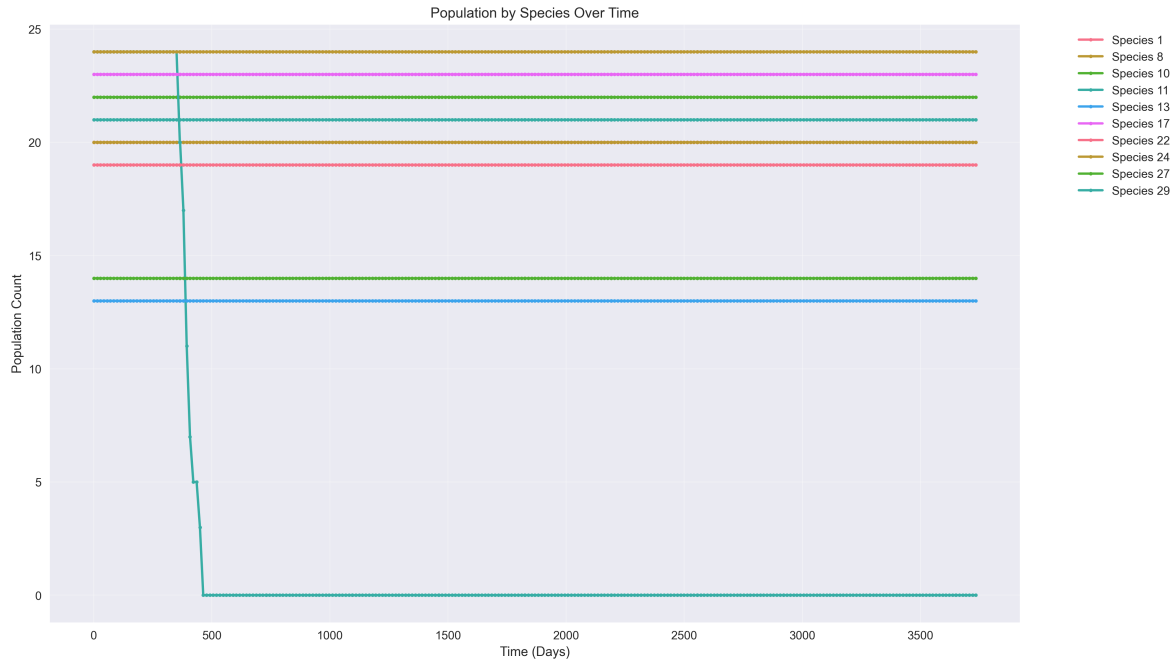


Figure 14: Graph of population over time from test 1

Test 1 was meant to establish a baseline for the ecosystem. From the graph above, it can be seen that the seeding function was malfunctioning, since plant reproduction did not occur.

Test 2

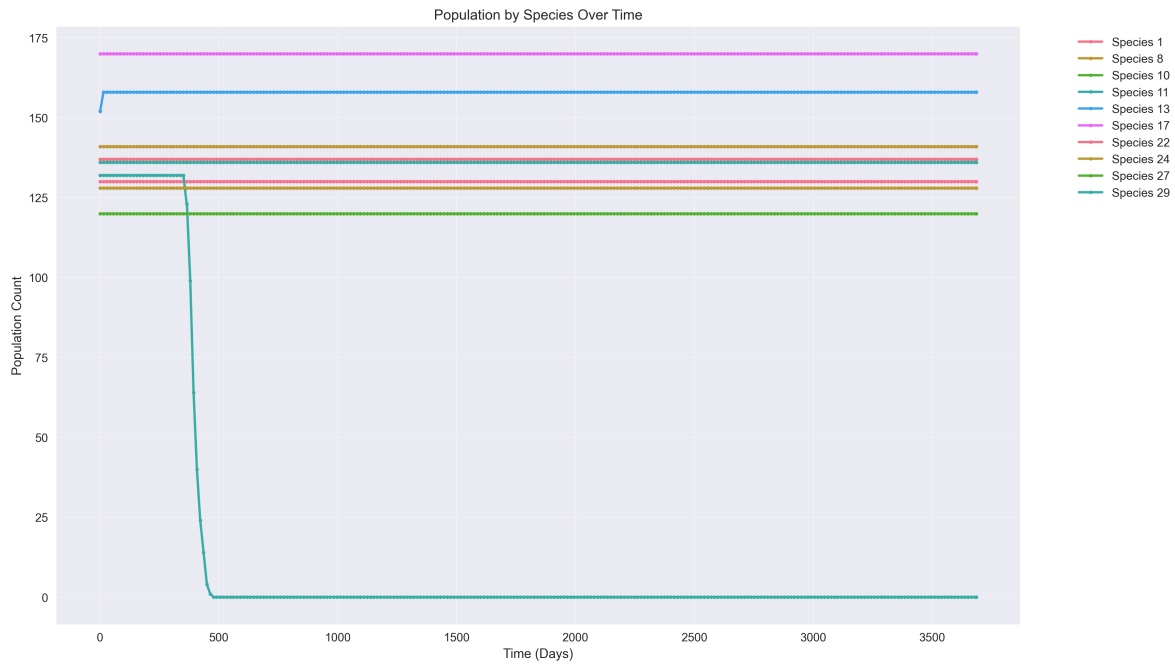


Figure 15: Graph of population over time from test 2

Test 2 started out with more plants, with some background changes for resource regeneration (increasing the overall rate). This test was used to rule out if the lack of resources was the cause. From the graph above we concluded that it wasn't, and the log snippet in figure 16 further shows the same.

Final Resource Levels (Average):

Water: 43.28

Nitrogen: 100.00

Phosphorous: 100.00

Figure 16: Data from simulation_summary_report from test 2

Test 3

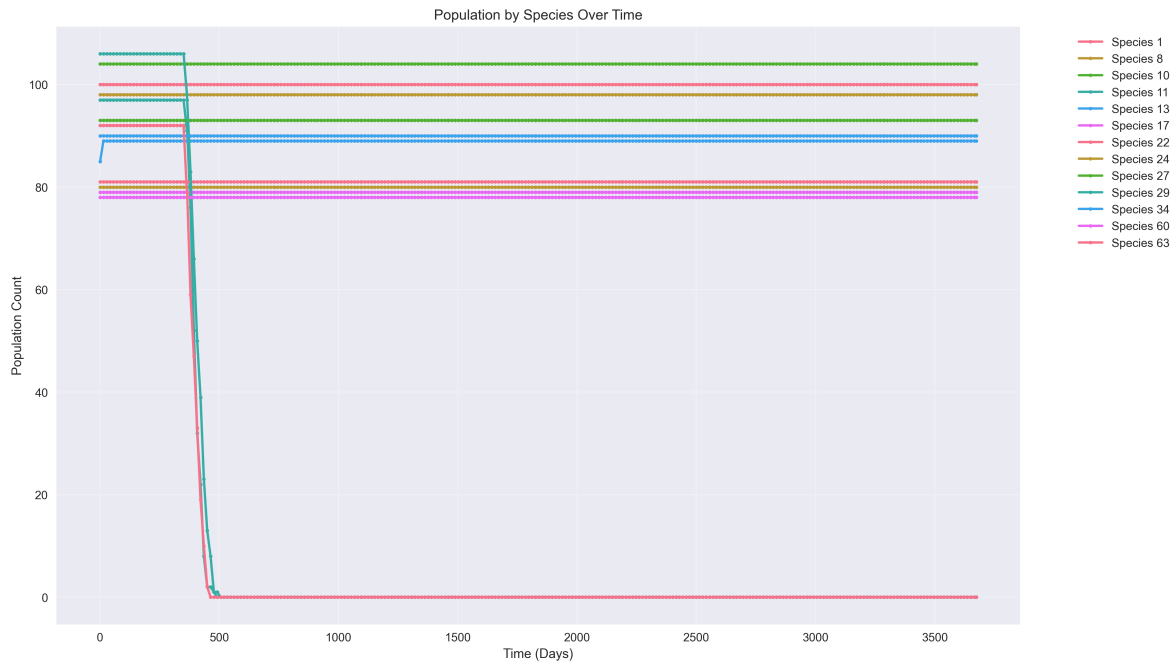


Figure 17: Graph of population over time from test 3

Test 3 introduced the invasive species to the simulation, but the lack of reproduction was still not fixed.

We temporarily disabled the checks for germination, but as it turns out, no plants even got to the flowering stage as shown in the following graph of population by growth stage over time:

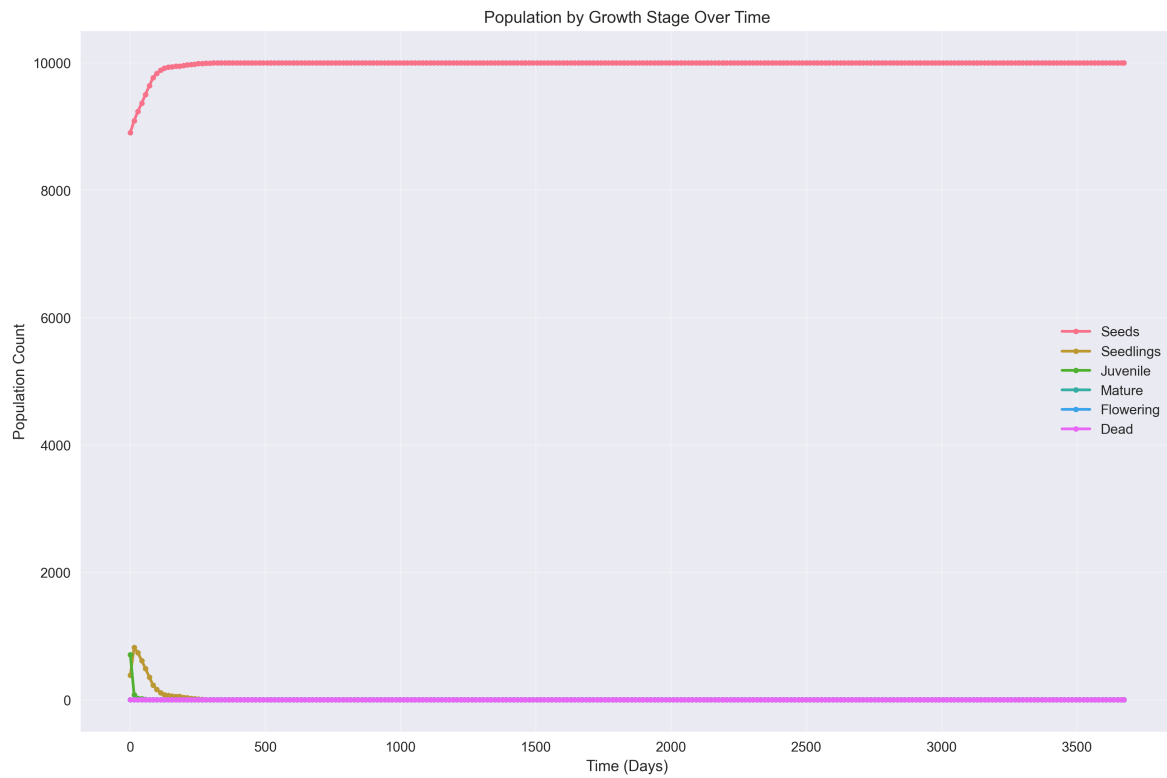


Figure 18: Graph of stages over time from test 3

Test 4

For the last test we had time to do, Test 4, we decided to remove the part that decreases the plant stage when there are insufficient resources. Surprisingly, this fixed the issue and made the reproduction aspect function, as can be seen in the graph above.

Strangely, this only affected the invasive species with no other clear explanation.

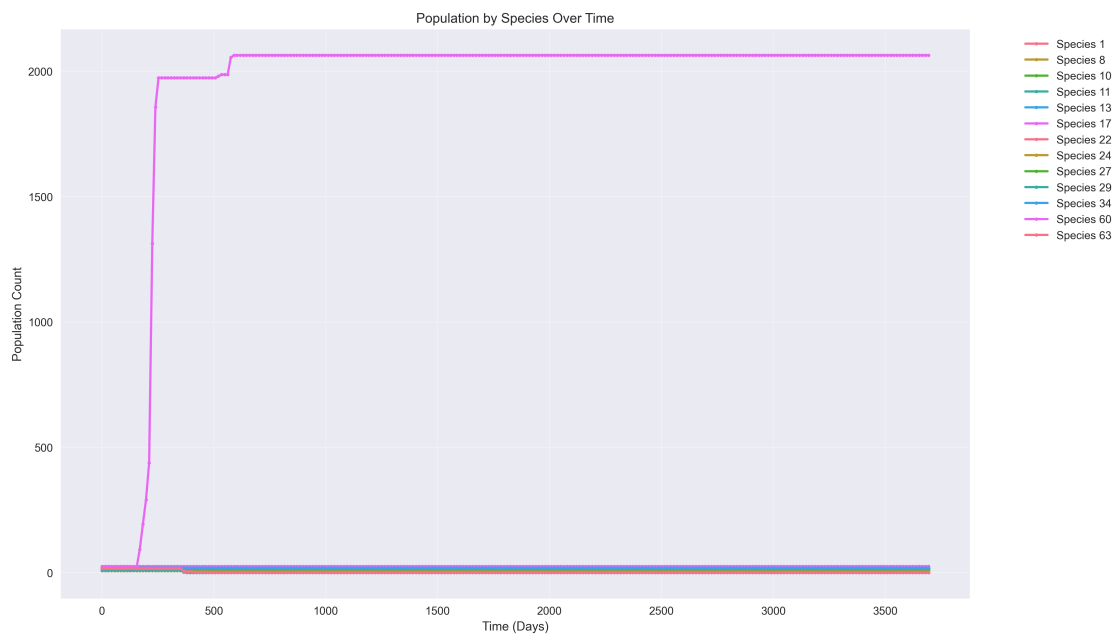


Figure 19: Graph of population over time from test 4

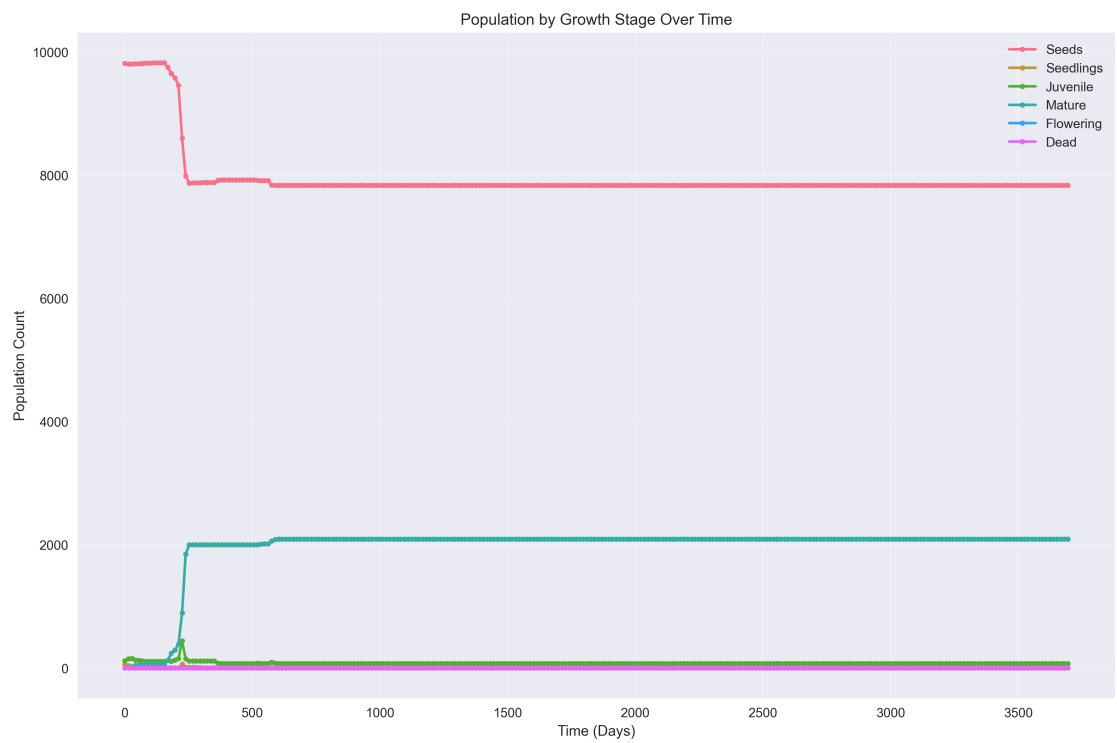


Figure 20: Graph of stages over time from test 4

6 Discussion

6.1 Possible improvements

Our current model still lacks some crucial ecological processes. Herbivores, pollinators, and other interactions that go beyond resource competition are not included. While adding these interactions would make the model more complex, it would also make it more realistic. Our cellular automata, unlike matrix models that assumes homogeneous environments, captures the landscape heterogeneity that drives invasion patterns. Compared to analytical invasion models that predict average spread rates, our approach reveals the spatial variability and stochastic elements that determine actual invasion trajectories.

The simulation so far demonstrates reasonable resemblance with observed patterns at Rødneæs, particularly in reproducing the spatial clustering characteristics of *Rosa Rugosa*'s invasions and the persistence of native species in resource-rich patches. Additional discrepancies between simulated and observed patterns can be attributed to the potential missing processes. The model tends to underestimate the speed of *Rosa Rugosa* establishment in the ecosystem, which could suggest that our animal dispersal algorithm may not fully capture the role of long distance seed transport.

The 2 week timestep scale we chose is a response to other studies that used a one year timestep. While using such a small timestep is more demanding computationally, it allows us to see how the environment got to that point. This scale is a baseline for the logic of the program, since data can be recorded at any interval we specify. Smaller timestep scales could work, however, since we are working with plants, most of said timesteps would be uneventful.

Arbitrarily assigning consumption rate and needs is more of a practical measure, as researching all the plant species to determine objectively how much they use could be possible, but it runs into the issue of wasting time on a small part of the simulation. We are looking at plant interactions, the soil resources allow said interactions to happen, but are a small part of a larger system.

The program runs based on states (0 = seed, 1 = seedling, 2 = juvenile, 3 = mature plant, 4 = flowering plant, 5 = dead plant). Such a system does leaves room for an evolutionary system, but it is not ideal. Realistically speaking, in the span of 10 years, major evolutions are quite unlikely to happen and representing that in data form would add unnecessary complexity. Climate change could be added, although its effects wouldn't be too noticeable for the same reason.

Such a model can be applied for conservation applications, although our implementation is flawed in many ways, as described in the previous chapter. Despite these limitations, our plant ecosystem simulation successfully captures many key features of real plant communities. The combination of realistic ecological processes, spatial structure, and quantitative output makes it a valuable tool for both basic research and applied conservation questions.

Our choice of grid resolution currently captures resource heterogeneity and local competitive interactions but may miss important landscape-level processes. Seed dispersal events that occur across multiple kilometers cannot be adequately represented. The models' local interaction rules may not scale appropriately to different resolutions.

7 Conclusion

This report presents a spatially explicit cellular automata model for simulating plant community dynamics and invasive species spread, using *Rosa Rugosa* as our invasive species. By integrating resource competition, various realistic dispersal mechanisms, and life cycle dynamics, our model captures key ecological processes that drive invasion patterns in ecosystems. The technical implementation demonstrated that complex ecological interactions can be represented through computational rules. Through the use of NumPy arrays, we could create a flexible framework capable of handling 12 species.

Our success with *Rosa rugosa* in Danish grasslands establishes cellular automata as a viable alternative to traditional invasion modelling approaches. The framework's transparency, spatial explicitness, and computational efficiency offer distinct advantages over the more complex differential equation systems. The approach is readily transferable to other invasive species and grassland types, requiring parameter adjustment rather than fundamental restructuring.

To answer our research question, "Is it possible to model the spread and resilience of invasive species on a specific type of grassland using cellular automata?", the short answer would be yes. Through our report, we successfully bridge the gap between theoretical ecology and real life, offering quantitative tools for conservation strategies.

However, the implementation did not get as far as we wanted in this project, since we wanted to code an entire plant species ecosystem, with many different dynamics, as detailed earlier in the report. However, we met with struggles with the coding aspects, and our final simulation is far from the complex system we wanted to model.

We believe that this project is possible to complete with a more cohesive implementation. It is a time-consuming project, with many little details to take into account, since ecosystems are full of complex dynamics. We believe that it is theoretically possible to model an ecosystem and its resilience to invasive species, together with the spread of species as well. These simulations could be set up to run for long periods of time and if done correctly, could potentially help us predict how different species will survive and spread. This could prove valuable in ecological studies of how to protect native ecosystems for years to come.

8 Use of GAI

During the preparation of this project report, we used the following tools that are based on generative AI: ChatGPT and Magic Media.

1. Use of GAI in the project process

Have you used GAI to search for information?

No

Have you used GAI to receive feedback and criticism on text?

Yes

Have you used GAI for proofreading?

No

Have you used GAI to improve the language of the text?

No

Have you used GAI to get ideas and new perspectives?

No

Have you used GAI to suggest a research question?

No

2. Use of GAI in the project report

Have you used GAI to figure out which sections the report should contain and what should be in each section?

No

Have you used GAI to write texts in the report?

No

Have you used GAI to make an abstract for the report (remember to cite the source in the relevant sections of the project report)?

No

Have you used GAI to find out how your empirical data should be analyzed?

No

Have you used GAI to carry out analyses of your or other people's empirical data or data?

No

Have you used GAI to generate figures?

No

Have you used GAI to generate code?

Yes, mostly used to fix bugs.

3. Have you used GAI in another way than the above?

If yes, describe where and how.

No

General questions

After using the GAI tools, the content was reviewed and edited. The authors take full responsibility for the content of this report.

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A Appendix

Source 1: The excel sheet that contains all the plant data we used.

URL: <https://zenodo.org/records/15522471>^[9]

Source 2: Contains the graphs and code base we used (version 2).

URL: <https://zenodo.org/records/15522533>^[10]